

ARM PrimeCell

Synchronous Static Memory Controller (PL093)

Revision: r0p3-00rel0

Release Note



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Release Information

Change history		
Issue	Date	Change
A	10 June 2003	First release.
B	25 February 2003	Updated for release r0p1-00ltd0
C	20 May 2003	Updated for patch release r0p1-00ltd1
D	22 Oct. 2003	Updated for patch release r0p1-00ltd2
E	12 Dec 2003	Updated for release r0p2-00bet1
F	4 March 2004	Updated for release r0p2-00rel0
G	27 September 2004	Updated for release r0p3-00rel0

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- *the document title*
- *the document number*
- *the page number(s) to which your comments refer*
- *a concise explanation of your comments*

General suggestion for additions and improvements are also welcome.

Feedback on the ARM PrimeCell Synchronous Static Memory Controller

If you have any comments or suggestions about this product, please contact your supplier giving:

- *the Product name*
- *a concise explanation of your comments.*

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1 PRODUCT DELIVERABLES

1.1 ARM Part Numbers for this product

The ARM part numbers for the individual deliverables that form part of the delivery for the ARM PrimeCell *Synchronous Static Memory Controller* (SSMC) peripheral are listed in [Table 1-1](#):

Table 1-1: ARM part numbers for ARM PrimeCell SSMC PL093

ARM PrimeCell PL093 Product deliverables	Document number	Product part number
Synthesizable VHDL RTL	N/A	PL093-MN-23001-r0p3-00rel0
Synthesizable Verilog RTL	N/A	PL093-MN-22001-r0p3-00rel0
Synthesis Scripts (Synopsys Design Compiler, Avant! CB18 cell library)	N/A	PL093-MN-01002-r0p3-00rel0
Testbench - Functional Verification, VHDL	N/A	PL093-MN-23010-r0p3-00rel0
Testbench - EASY Integration test, VHDL	N/A	PL093-MN-23012-r0p3-00rel0
Testbench - Functional Verification, Verilog	N/A	PL093-MN-22010-r0p3-00rel0
Testbench - EASY Integration test, Verilog	N/A	PL093-MN-22012-r0p3-00rel0
Test vectors - BusTalk, functional verification	N/A	PL093-VE-01010-r0p3-00rel0
Test vectors - BusTalk, integration test	N/A	PL093-VE-01012-r0p3-00rel0
ARM PrimeCell SSMC (PL093) Technical Reference Manual (PDF)	ARM DDI 0236	PL093-DA-03001-r0p3-00rel0
ARM PrimeCell SSMC (PL093) Design Manual (PDF)	PL093 DDES 0000	PL093-DC-02003-r0p3-00rel0
ARM PrimeCell SSMC (PL093) Integration Manual (PDF)	PL093 INTM 0000	PL093-DC-10002-r0p3-00rel0
ARM PrimeCell SSMC (PL093) Release Note (PDF)	PL093 RELN 0000	PL093-DC-06002-r0p3-00rel0
ARM PrimeCell SSMC (PL093) User Guide (PDF)	PL093 USGD 0000	PL093-DC-08001-r0p3-00rel0
Denali Memory Models	N/A	PL093-MN-28001-r0p3-00rel0
The parts with numbers starting PL000 are delivered as a separate bundle PL000-BU-00004		
System Common and System Indirection Source Code	N/A	PL000-SW-02001-r9p0-00rel0
System Test Source Code	N/A	PL000-SW-02003-r9p0-00rel0
System Common Files Source Code	N/A	PL000-SW-02004-r9p0-00rel0
System Build Project Files	N/A	PL000-SW-02005-r9p0-00rel0
Interrupt Controller Object Code	N/A	PL001-SW-00000-r9p0-00rel0
Timer Object Code	N/A	PL002-SW-02001-r9p0-00rel0
Software Code Module	N/A	PL093-SW-02001-r0p3-00rel0
Software Test Report	PL093_TREP_1000	PL093-DC-01001-r0p3-00rel0
Software API Document	PL093_ISPC_1000	PL093-DC-02001-r0p3-00rel0
System Structure and Integration Guide	PL000_DII_1000	PL000-DC-02000-r9p0-00rel0
System Common API Documents	PL000_ISPC_1001	PL000-DC-02001-r9p0-00rel0
System Indirection API Documents	PL000_ISPC_1002	PL000-DC-02001-r9p0-00rel0

System Test API Documents	PL000_ISPC_1003	PL000-DC-02003-r9p0-00rel0
System Common Files API Documents	PL000_ISPC_1004	PL000-DC-02004-r9p0-00rel0
Interrupt Controller API Documents	PL001_ISPC_1000	PL001-DC-02001-r9p0-00rel0
Timer API document	PL002_ISPC_1000	PL002-DC-02001-r9p0-00rel0

The following part is delivered separately and no longer within the main product bundle

ARM PrimeCell Synchronous Static Memory Controller (PL093) Errata Notice (PDF)	PL093 DEI 0000	PL093-DC-11001
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In addition to the above deliverables a reference cell library is also included (see [Table 1-2](#)) to allow the peripheral synthesis to be carried out.

Table 1-2: Reference cell library bundle (PL000-BU-00005-0.00)

Part number	Description
PL000-BU-00005-0.00	Synopsys CB18 (0.18um) reference cell library - Synopsys and Verilog views

NOTE The part numbers in the above table are not distinguishable parts in the installed directory structure and are only defined in this way for contractual purposes.

2 INSTALLATION

2.1 Introduction

Intellectual Property (IP) deliverables are collectively delivered as a single UNIX compressed tar file which has been security encrypted.

An ARM PrimeCell peripheral may be one of several products delivered as part of this single encrypted file. Each ARM PrimeCell peripheral comprises a set of compressed tar files which correspond to a unique part number quoted in the contractual agreement document. A corresponding list file details the files in each tar file

These installation instructions relate to the Unix platform/operating system.

2.2 Disk space required

This installation will require 87 Mbytes of free disk space.

2.3 Installation procedure

You will have a single tarred, zipped and DES encrypted file which will have the following typical filename format:

`<company_name>-arm-download-<transaction_ID>.tgz.des`

where `<transaction_ID>` is a unique 5-digit transaction ID number.

The installation procedure is summarised below:

2.3.1 Unpacking the shipment

The following steps describe how to unpack the deliverables in this shipment.

1. Relocate the encrypted file (for example `<file>.tgz.des`, if DES encryption has been used) to an appropriate location that has sufficient free disk space.
2. Follow the decryption instructions received in the notification e-mail to generate a decrypted, compressed, tar file `<file>.tgz`
3. Unzip and untar the file by the use of `gunzip` and `gtar` commands to extract the deliverables as follows:

```
gunzip <file>.tgz
gtar -xvf <file>.tar
```

NOTE It is recommended that `gtar v1.13.25` (or later) be used to unpack the deliverables as other versions of `tar` do not always handle long directory paths correctly. To find out the version of `gtar` installed type `gtar --version`

The following files and directories will be created:

ARM_MANIFEST_<transaction_ID>.TXT	(a complete list of the files delivered)
ARM_DELIVERY_<transaction_ID>.TXT	(lists the ARM bundles and parts delivered)
PL093-BU-000000-r0p3-00rel0/PL093_VC/...	(PL093 virtual component files)
PL093-BU-000000-r0p3-00rel0/PL093_SW/...	(PrimeCell PL093 software driver files)
PL000-BU-000005-0.00/Avanti	(reference cell library files)
PL000-BU-000004-r9p0-00rel0/PrimeCell_gen_sw	(PrimeCell common driver files)
PL093-DC-11001-r0p3-00rel0/...	(PrimeCell PL093 Errata Notice)

NOTE: It is recommended that `gtar v1.13.25` (or later) be used to unpack the deliverables as other versions of `tar` do not always handle long directory paths correctly. To find out the version of `gtar` installed type `gtar --version`

2.3.2 Installation of the reference cell library

The reference cell library bundle will be included in the `arm-download-<transaction_ID>.tgz` file if the PL093 is the only peripheral in the delivery. When the `arm-download-<transaction_ID>.tgz` file is unpacked following the process in section 2.3.1, there will appear a directory named:

`PL000-BU-000005-0.00/Avanti`

This contains a tar file named `CB18_2000.3.tar.gz` which contains the reference cell library Synopsys and Verilog cell views.

To setup the cell library in a user's directory, for example `/ARM/libraries`, copy the tar file to the user's directory:

```
cd /ARM/libraries
cp /PL000-BU-000005-0.00/Avanti/CB18_2000.3.tar.gz .
```

Execute the UNIX `gunzip` and `gtar` commands to extract the cell library files as follows:

```
gunzip CB18_2000.3.tar.gz
gtar -xvf CB18_2000.3.tar
```

The *PL093 User Guide* in the `ssmc_pl093/docs` directory details how to setup the peripheral development environment to use the reference cell library and describes the reference cell library directory structure. The User Guide also gives instructions for running testbench simulations and synthesis on the deliverables supplied.

2.3.3 PrimeCell common driver files

The installed directory PL000-BU-00004-r9p0-00rel0/PrimeCell_gen_sw contains the PrimeCell common driver files required for using the supplied PL093 software driver deliverables in the structure illustrated in [Figure 2-1](#).

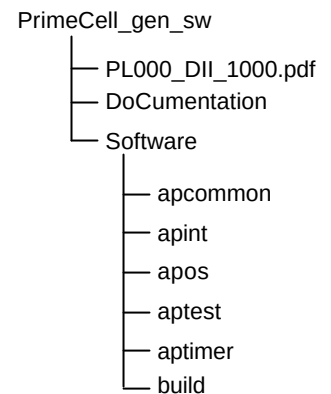


Figure 2-1: PrimeCell common driver files directory structure

The *ARM PrimeCell Drivers System Integration Guide* (PL000_DII_1000 . pdf) describes how the PrimeCell driver can be integrated into a system.

NOTE: The PL093 software driver and PrimeCell common driver files are not required for running the PL093 virtual component testbench simulations and are supplied as an example driver.

The DoCumentation contains specifications on the PrimeCell driver framework APIs.

3 DOCUMENTATION

3.1 Adobe PDF

Documents are supplied as “Adobe PDF” (Portable Document Format) files. These files are readable on most common computer platforms and operating systems using Adobe Acrobat Reader. The Adobe Acrobat Reader software can be downloaded from the Adobe World Wide Web site at <http://www.adobe.com>.

3.2 Peripheral Documents Supplied

The documents supplied with the ARM Synchronous Static Memory Controller PL093 are located in the `ssmc_pl093/docs` directory and are listed in [Table 3-1](#).

Table 3-1: List of PL093 documents supplied

Document Title	Document Number	Description of content
PL093 SSMC Technical Reference Manual	ARM-DDI-0236	Specification describing the peripheral functionality, programmers model and signal I/O.
PL093 SSMC User Guide	PL093-USGD-0000	Overview of the PL093 peripheral development environment, directory structure and files. Instructions are given on how to run simulation and synthesis scripts and describes what output files are produced.
PL093 SSMC Integration Manual	PL093-INTM-0000	Guide to integrating the peripheral in to a system and covers topics such as synthesis, clocks/resets, pad interfacing, RTL customization, etc.
PL093 SSMC Design Manual	PL093-DDES-0000	Describes the design implementation and gives details of the functional verification testbenches and tests.

4 PRODUCT STATUS

4.1 Product Maturity

This product is newly developed IP and is initially released with Early Access status.

Early Access release status has a particular meaning to ARM of which the recipient must be aware. A product so designated has satisfactorily achieved all criteria for its promotion to Full Release status. It may be delivered in accordance with the contract and be expected to perform as described in the data-sheet. However, there remains some elements of uncertainty, which cannot finally be validated until the product has been successfully deployed by customers. Accordingly, the recipient of a product with Early Access-Release status, may be directly contributing to the final stage of approval of that product.

In due course, the Early Access-Release will be replaced by the Full-Released version or where no changes have been required to the product's deliverables a simple notification will be given that the product has now been given Full Release status. This release note will not necessarily be updated to indicate a move to Full Release. It should be noted that Support for the Early Access release of the product will only be provided by ARM to a recipient who has a current support and maintenance contract for the product.

4.2 Summary synthesis results

[Table 4-1](#) gives a summary of the synthesis results of the ARM SSMC PL093 using Avant! Passport 2000.3 CB18 v1.0 (0.18um) cell library.

Table 4-1: Summary synthesis results

HDL Source		VHDL	Verilog
Area (without scan)	Total cell area	20413	19530
	Net interconnect area	7790	7487
Area (scan inserted)	Total cell area	24635	24804
	Net interconnect area	9207	9239
Target Clock frequency		133MHz	
Scan insertion figure % (number of 'D' flip-flops replaced)		100% (1303)	100% (1303)
Estimated scan fault coverage		99.12%	98.92%
Synopsys tool versions		Design Compiler and Test Compiler 2003.06-SP1	

4.3 Design tools used during release testing

[Table 4-2](#) lists the design tool versions used in the testing of this release.

Table 4-2: Design tool versions

Tool category	Vendor tool and version
Synthesis and scan test	Synopsys Design Compiler and Test Compiler 2003.06-SP1
Simulation	Model Technology Inc, ModelSim 5.5d Cadence Verilog-XL LDV3.2 DenaliSoft Inc, Denali Memory Modeller AV 3.100-0050
ARM Global Peripheral Development Environment	ARM Release REL1v7
Miscellaneous	gcc 2.95.3 (required for compiling BusTalk verification vectors) gtar 1.13.25 (recommended for unpacking deliverables)

4.4 BusTalk functional block verification conditions

Functional block verification has been performed with the following conditions using the BusTalk testbench with free-running clocks.

SSMCCLK/PCLK	Simulations
133MHz (7.5ns period)	VHDL and Verilog RTL, Verilog only gate-level min/max SDF

For further details of functional verification testing, please refer to the *ARM PrimeCell Synchronous Static Memory Controller (PL093) Design Manual*.

5 DIFFERENCES FROM PREVIOUS REVISIONS

5.1 Differences between r0p2-00rel0 and r0p3-00rel0

The following defects have been corrected. Please refer to the Errata Notice for more details:

- ✚ 326300: Write data that appears to have been accepted may not be written to memory
- ✚ 322167: AhbWideRdCounter is Being Incorrectly Decrementd
- ✚ 323746: Performance Inefficiency When Synchronous Burst Not Started on Burst Boundary
- ✚ 331175: Read data loss during page mode reads

6 KNOWN ISSUES

Please refer to the Errata Notice for more details about the item below:

- ✚ 150012: System performance is not optimum if HSIZE is not equal to the Memory Width.

Any issues will be documented in new versions of the Errata Notice. Please refer to the Errata Notice document in this release for more details of past errata issues which have been fixed in this release.